

I picked Logistic Regression and k-Nearest Neighbors (KNN) for this run because they look at classification from opposite sides of the room. Logistic Regression is parametric, meaning it sets up a mathematical line—or a hyperplane once you stack multiple features together—and runs that linear combination through an S-shaped sigmoid curve to spit out a clean probability between 0 and 1. If that probability crosses your decision threshold, the patient gets flagged as diabetic. KNN skips formulas altogether. It maps every patient into multi-dimensional space like points on a scatter plot and checks who is standing nearby. When a new patient record comes in, the nearest neighbors basically take a vote on the diagnosis. One relies on a global mathematical equation, while the other just follows the local crowd.

I split the data using an 80/20 train-test ratio, tossed in `stratify=y` to keep the ratio of diabetic to non-diabetic cases even on both sides, and set `random state=42` so the numbers stay fixed every time the cell runs. For Logistic Regression, I made three tweaks past the defaults: I set `solver='liblinear'` because it handles smaller tabular sets without fussing, increased `max_iter=1000` to make sure the solver has plenty of room to converge, and dropped the regularization parameter down to `C=0.5` to keep the weights from overfitting. For KNN, I bumped `n_neighbors` from 5 to 7 to smooth out random quirks and avoid tied votes, switched to `weights='distance'` so closer data points carry more weight than distant neighbors, and set `metric='manhattan'` because city-block distance handles outliers better across mixed clinical measurements than standard straight-line distance.

Without running any evaluation metrics yet, my bet is on Logistic Regression taking the lead here. Health markers like glucose levels, BMI, and age tend to have a fairly direct, compounding relationship with diabetes risk—meaning as those numbers climb, the chances of diabetes generally trend up too. Since the prompt mentioned the data has already been cleaned and scaled, a straightforward linear boundary should catch those overarching trends pretty reliably. KNN can be great, but in a multi-dimensional space where clinical numbers overlap and borders get messy, relying on nearby neighbors can easily get thrown off by a few borderline records.

Outside of healthcare, a clear example of binary classification in industry is how Netflix forecasts subscriber churn [1]. Their models evaluate activity data—like weekly streaming hours, browsing drop-offs, and how often a profile goes dormant—to classify whether an account is likely to cancel its subscription next month [1]. Once an account gets flagged in that high-risk bracket, the system triggers automated retention workflows, pushing tailored title recommendations or re-engagement emails to win the viewer back before they unsubscribe [1].

Reference

[1] AlmaBetter, "Netflix Churn Rate Prediction Case Study," *AlmaBetter Insights*, Feb. 2024.
[Online]. Available: <https://www.almabetter.com/bytes/articles/netflix-churn-prediction-case-study>.